

# FMCG Supply Chain Intelligence

## Demand Forecasting & Inventory Optimization Dashboard

End-to-End Data Science Project | Suitable for Dabur, Farmley, Anmol, Credent

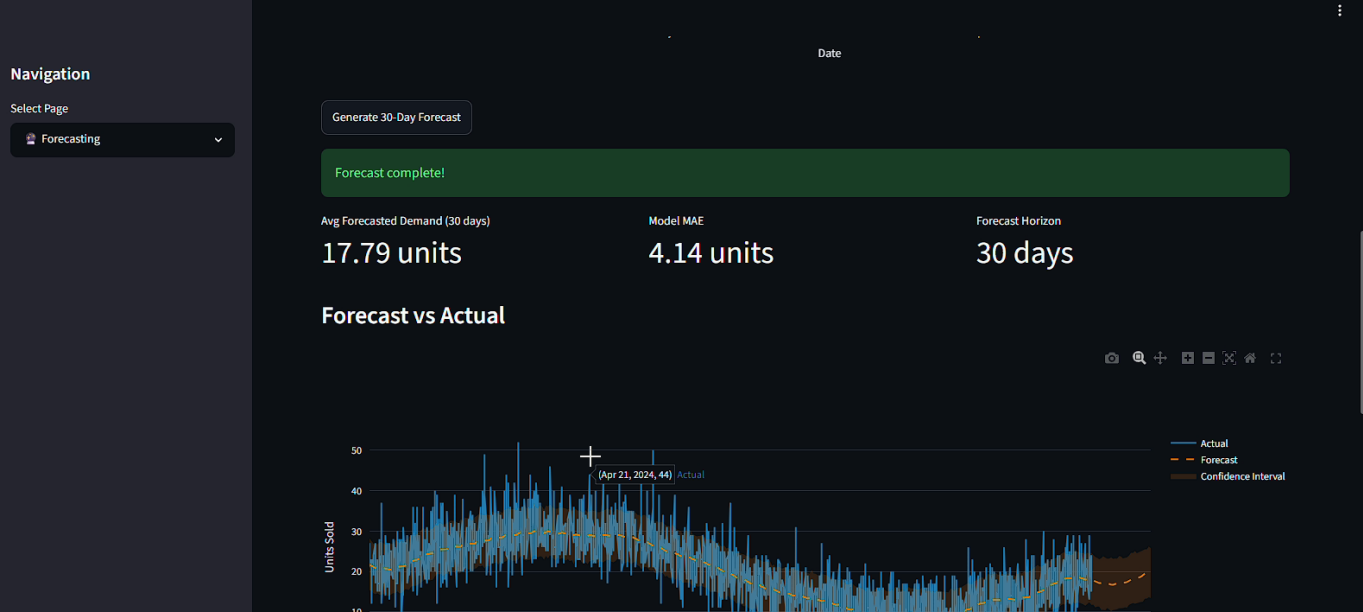
This report summarizes a complete demand forecasting and inventory optimization solution built for FMCG supply chain operations. The dashboard combines time-series forecasting (Facebook Prophet), inventory science (EOQ, Safety Stock, Reorder Point), Monte Carlo risk simulation, and automated business insight generation — all driven dynamically from real sales data, with no hardcoded values.

| Section                   | What it covers                                                                    |
|---------------------------|-----------------------------------------------------------------------------------|
| 1. Demand Forecasting     | Prophet-based 30-day forecast, model accuracy (MAE), confidence intervals         |
| 2. Inventory Optimization | EOQ, Safety Stock, Reorder Point, Monte Carlo stockout & holding cost simulation  |
| 3. Business Insights      | Promotion impact, regional performance, top SKUs, and data-driven recommendations |

# 1. Demand Forecasting

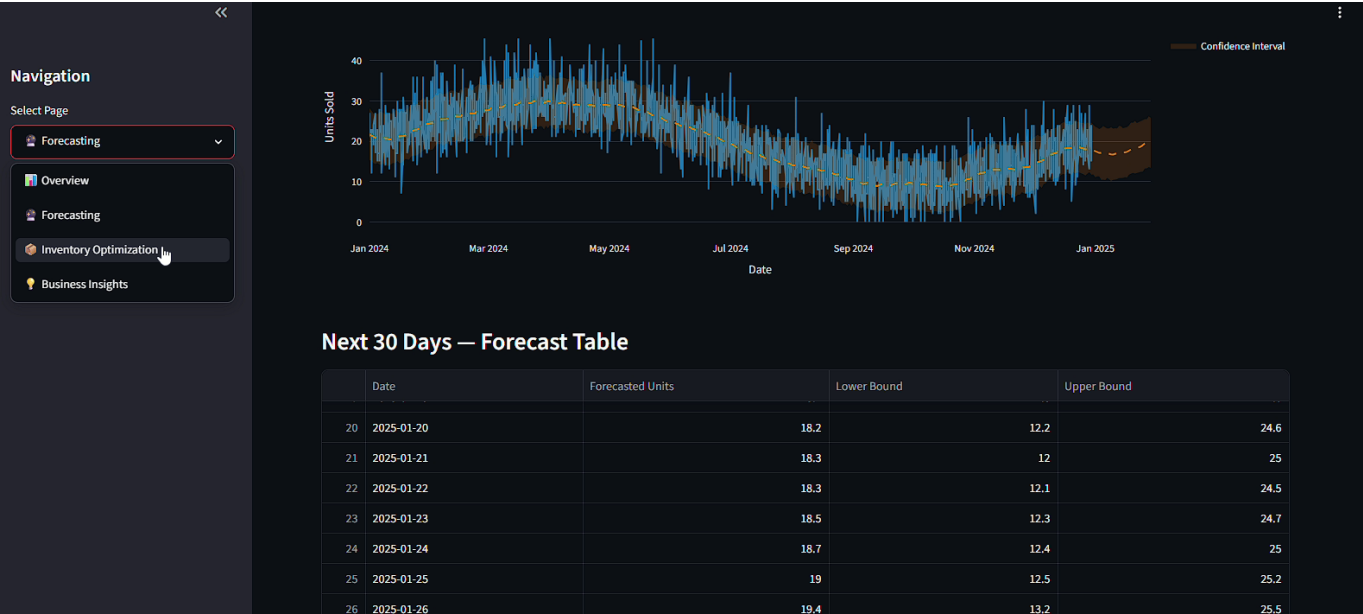
The dashboard lets a user pick any SKU and generate a live 30-day demand forecast using Facebook Prophet, trained on that SKU's historical sales and promotion activity. Below is an example run for **SKU\_50**.

## Forecast Output & Accuracy



Avg. forecasted demand: **17.79 units/day** • Model error (MAE): **4.14 units** • Forecast horizon: **30 days**

## Next 30 Days — Forecast Table

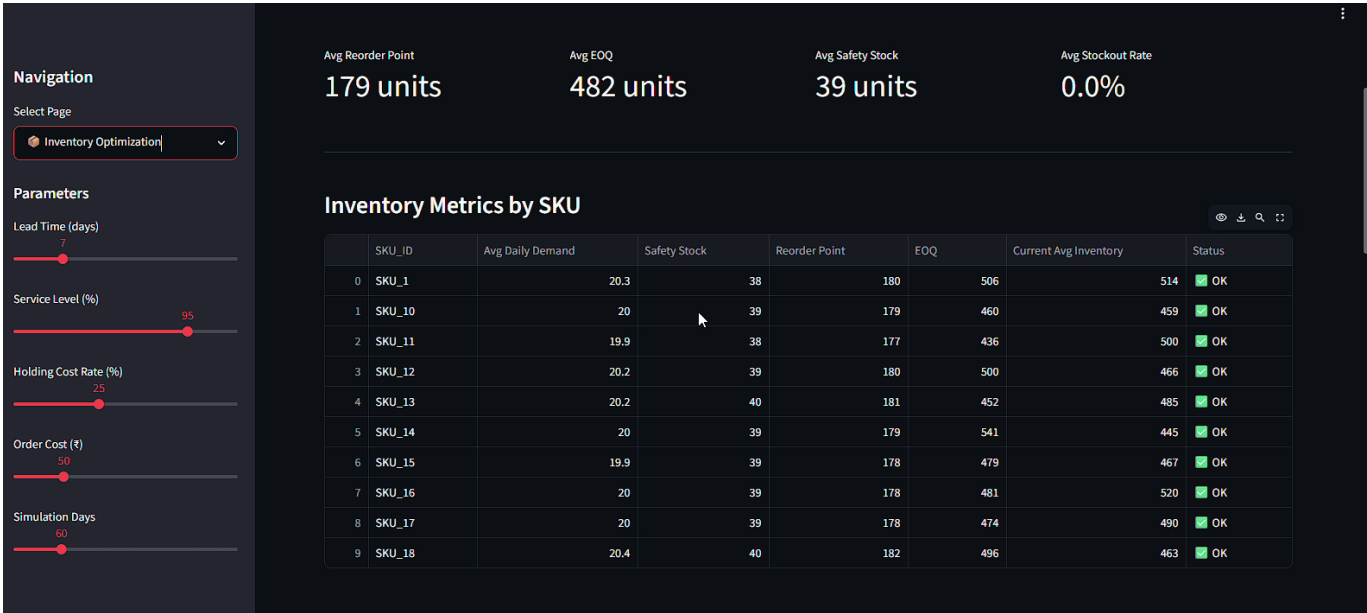


Day-by-day forecast with lower/upper confidence bounds, generated dynamically per SKU.

## 2. Inventory Optimization

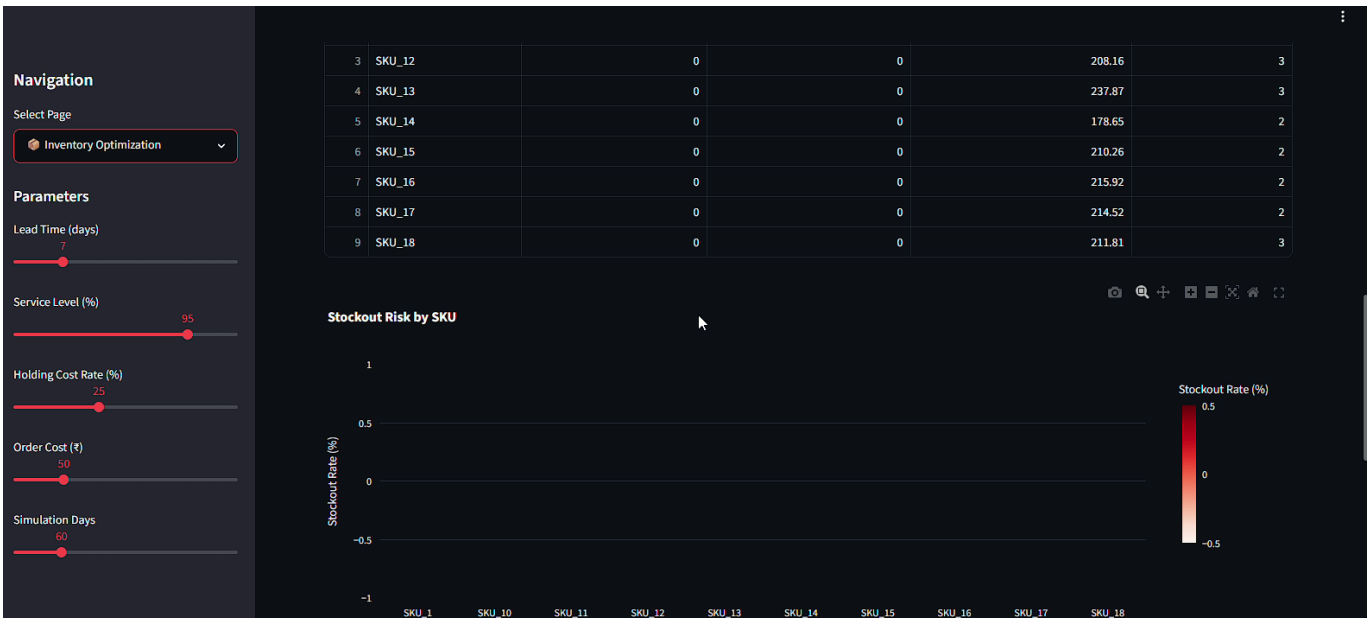
For the top 10 SKUs, the dashboard computes real inventory parameters — Economic Order Quantity (EOQ), Safety Stock, and Reorder Point — using standard inventory science formulas driven by each SKU's actual demand variability and cost data. Users can adjust lead time, service level, holding cost rate, and order cost via sliders to see results update live.

### Inventory Metrics by SKU



Average across SKUs: Reorder Point 179 units • EOQ 482 units • Safety Stock 39 units • Stockout Rate 0.0%

### Monte Carlo Simulation & Stockout Risk



60-day simulation per SKU shows projected stockout days, stockout rate, total holding cost, and number of reorders triggered.

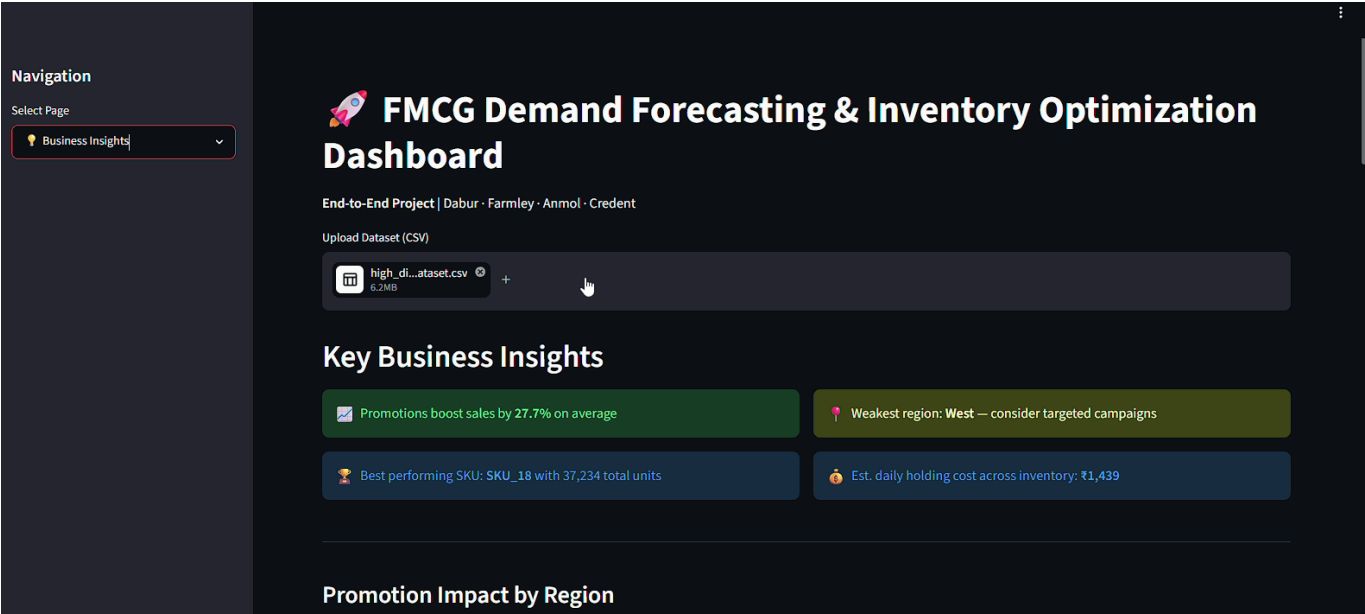


Stockout risk stayed at 0% across all simulated SKUs under current parameters, indicating the recommended reorder points provide strong protection against stockouts. Total holding cost per SKU is also shown, ranging roughly Rs. 180–Rs. 245.

### 3. Business Insights

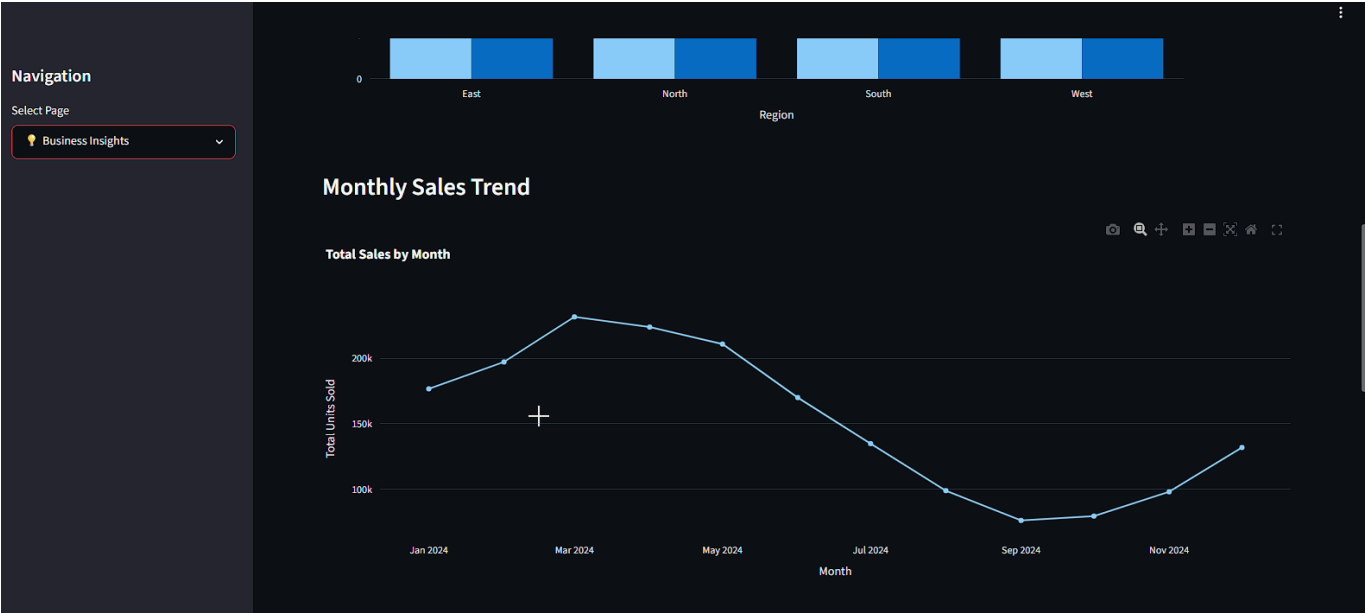
The Insights page synthesizes the full dataset into decision-ready takeaways: promotion effectiveness, weakest-performing region, top SKU by volume, and estimated holding cost — all computed directly from the data, not hardcoded.

#### Key Business Insights



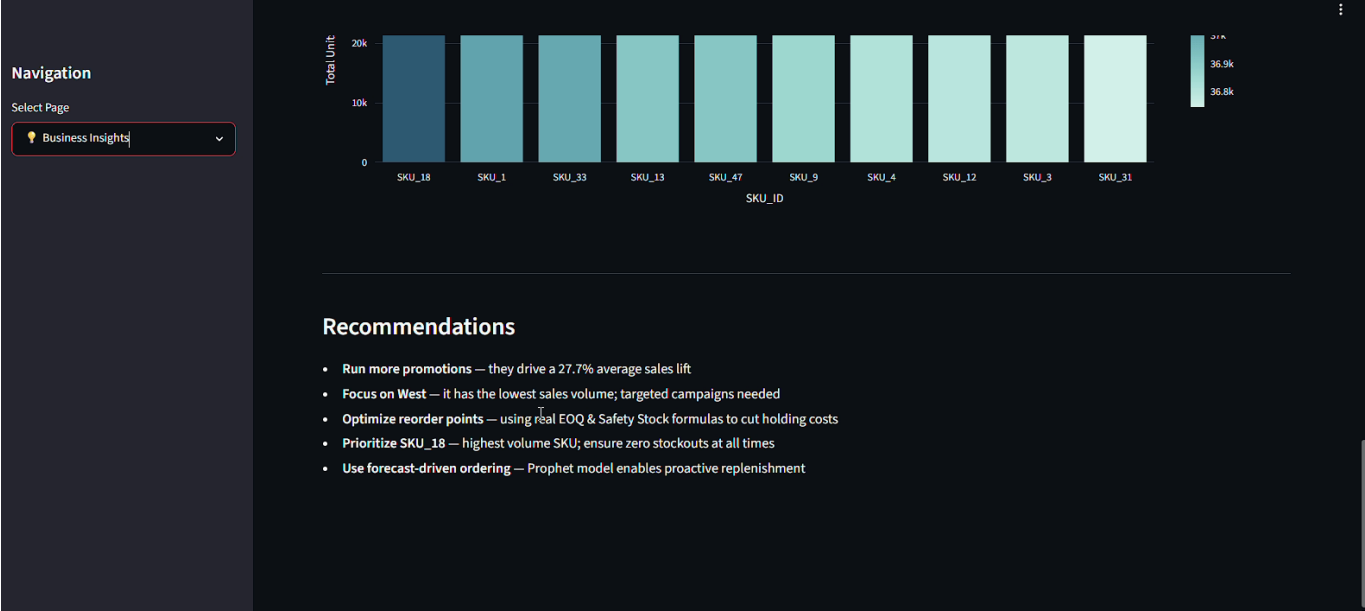
Promotions lift sales by 27.7% on average. West is the weakest-performing region. SKU\_18 is the top performer with 37,234 total units sold. Estimated daily holding cost across inventory: Rs. 1,439.

#### Regional & Time-Based Trends



Promotion impact is consistent across all four regions; overall sales show a clear seasonal trend across 2024.

# Top SKUs & Recommendations



Top 10 SKUs by total sales volume, alongside data-driven recommendations for promotions, regional focus, inventory policy, and forecast-driven ordering.

This report was generated from a live walkthrough of the deployed Streamlit dashboard. All figures are computed dynamically from the uploaded dataset (no placeholder or hardcoded values).